

Goal. Figure out if your infinite fence encloses a finite or infinite amount of area!

1. Evaluate $\int_3^6 \frac{x}{\sqrt{x-3}} dx$ using the following steps:

(a) Why is $\int_3^6 \frac{x}{\sqrt{x-3}} dx$ an improper integral?

(b) Rewrite the improper integral as the limit of a finite integral by replacing the lower bound with a variable a and taking a one-sided limit as $a \rightarrow 3^+$.

(c) Evaluate the finite integral.

(d) Now take the limit.

(e) State your conclusion: Does the integral converge or diverge? If it converges, what does it converge to? (That is, what is the finite area enclosed by your infinite fence?)

2. Use the same steps to determine for what values of p the following integrals converge:

(a) $\int_1^\infty \frac{1}{x^p} dx.$

(b) $\int_0^1 \frac{1}{x^p} dx.$

Comparison Theorem. Suppose $0 \leq f(x) \leq g(x)$ for all $x \geq a$ and f and g are continuous on $[a, \infty)$.

- If $\int_a^\infty g(x) \, dx$ converges, then $\int_a^\infty f(x) \, dx$ converges also.
- If $\int_a^\infty f(x) \, dx$ diverges, then $\int_a^\infty g(x) \, dx$ diverges also.
- In any other case, we know nothing!

3. Use the comparison theorem to determine whether the following improper integral converges or diverges: $\int_3^\infty \frac{5x}{(x^7 + x^2)^{1/3}} \, dx$.

(a) What is the approximate largest term in the numerator? What is the approximate largest term in the denominator? (Example: for $(x^2 + 1)^{3/2}$, we would say that the largest term is approximately $(x^2)^{3/2} = x^3$.)

(b) If we delete all terms in the integral except for the two largest terms in the previous part, does the integral converge or diverge? Why?

(c) Use the comparison theorem and part (b) to determine whether or not the original integral converges or diverges.

4. Use the comparison theorem to determine whether the following improper integral converges or diverges: $\int_1^{\infty} e^{-x^2} dx$.